



# Urine Toxicology Controls

Multi-analyte quality controls designed to monitor the precision of screening, semi-quantitative and confirmatory drugs-of-abuse test procedures

**BIO-RAD**



## Liquichek Urine Toxicology Screen Controls



### Liquichek Urine Toxicology Control, Levels S1E & S2E

A human urine control containing drugs of abuse at concentrations 20–25% above or below EIA screen cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
2,000 ng/mL  
Opiate cutoff



### Liquichek Urine Toxicology Control, Levels S1E Low Opiate & S2E Low Opiate

A human urine control containing drugs of abuse at concentrations 20–25% above or below EIA screen cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
300 ng/mL  
Opiate cutoff



### Liquichek Urine Toxicology Control, Levels S10 Low Opiate & S20 Low Opiate

A human urine control containing drugs of abuse at concentrations 20–50% below or above commonly used SAMHSA screen cutoff levels.

- 2 year shelf life at -20°C to -70°C
- 30 day open-vial stability at 2–8°C
- 45 day closed vial stability at 2–8°C
- Includes Methamphetamine and Oxazepam

▶ For use with  
300 ng/mL  
Opiate cutoff



## Which Screen Controls are Right for You?

Different methodologies, regulations and guidelines call for quality control materials spiked with purified drugs or drug metabolites at different concentrations. No matter how you define a standard drug panel, or what concentrations you need, Bio-Rad has a control for you.

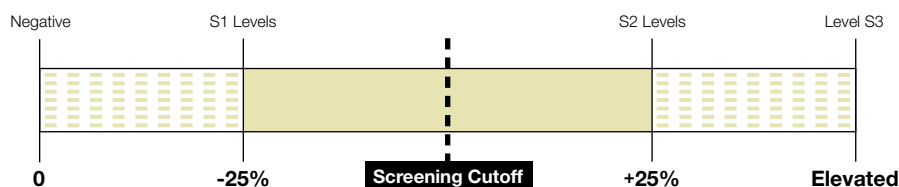
Bio-Rad Drugs-of-Abuse Controls are made from human urine with pH, specific gravity and creatinine that fall within the limits of normal human urine, and meet current SAMHSA\* and other agency guidelines.

### Liquichek Urine Toxicology Screen Control Gravimetric Reference Values

| Analytes (ng/mL)                  | Level S1E<br>Low Opiate | Level S2E<br>Low Opiate | Level S1E | Level S2E | Level S1<br>Low Opiate | Level S2<br>Low Opiate | Level S1 | Level S2 |
|-----------------------------------|-------------------------|-------------------------|-----------|-----------|------------------------|------------------------|----------|----------|
| Catalog Number                    | 423                     | 424                     | 438       | 439       | 466                    | 467                    | 461      | 462      |
| <b>Amphetamines</b>               |                         |                         |           |           |                        |                        |          |          |
| d-Amphetamine                     | –                       | –                       | –         | –         | 750                    | 1,250                  | 750      | 1,250    |
| d-Methamphetamine                 | 750                     | 1,250                   | 750       | 1,250     | –                      | –                      | –        | –        |
| MDMA                              | –                       | –                       | –         | –         | –                      | –                      | –        | –        |
| <b>Barbiturates</b>               |                         |                         |           |           |                        |                        |          |          |
| Secobarbital                      | 150                     | 375                     | 150       | 375       | 150                    | 375                    | 150      | 375      |
| <b>Benzodiazepines</b>            |                         |                         |           |           |                        |                        |          |          |
| Lormetazepam                      | 150                     | 375                     | 150       | 375       | –                      | –                      | –        | –        |
| Nordiazepam                       | –                       | –                       | –         | –         | 75                     | 375                    | 75       | 375      |
| Oxazepam                          | –                       | –                       | –         | –         | –                      | –                      | –        | –        |
| <b>Cannabinoids</b>               |                         |                         |           |           |                        |                        |          |          |
| 11-Nor-Δ <sup>9</sup> -THC-9-COOH | 37.5                    | 65                      | 37.5      | 65        | 37.5                   | 65                     | 37.5     | 65       |
| <b>Cocaine</b>                    |                         |                         |           |           |                        |                        |          |          |
| Benzoyllecgonine                  | 225                     | 375                     | 225       | 375       | 225                    | 375                    | 225      | 375      |
| Ethanol (mg/dL)                   | 40                      | 70                      | 40        | 70        | 40                     | 70                     | 40       | 70       |
| Lysergic Acid Diethylamide (LSD)  | 0.3                     | 0.8                     | 0.3       | 0.8       | 0.3                    | 0.8                    | 0.3      | 0.8      |
| Methadone                         | 225                     | 375                     | 225       | 375       | 225                    | 375                    | 225      | 375      |
| Methaqualone                      | 225                     | 375                     | 225       | 375       | 225                    | 375                    | 225      | 375      |
| <b>Opiates</b>                    |                         |                         |           |           |                        |                        |          |          |
| Morphine (Free)                   | 225                     | 375                     | 1,500     | 2,500     | 225                    | 375                    | 1,500    | 2,500    |
| Phencyclidine (PCP)               | 19                      | 31                      | 19        | 31        | 19                     | 31                     | 19       | 31       |
| Propoxyphene                      | 225                     | 375                     | 225       | 375       | 225                    | 375                    | 225      | 375      |
| <b>Tricyclic Antidepressants</b>  |                         |                         |           |           |                        |                        |          |          |
| Nortriptyline                     | 750                     | 1,250                   | 750       | 1,250     | 750                    | 1,250                  | 750      | 1,250    |

\*United States Substance Abuse and Mental Health Services Administration (formerly NIDA)  
Refer to the package insert for performance and stability claims.





| Level S3   | Level S1O<br>Low Opiate | Level S2O<br>Low Opiate | Level S1S  | Level S2S  | SAMHSA*<br>Screen<br>Cutoffs | Analytes (ng/mL)                 |
|------------|-------------------------|-------------------------|------------|------------|------------------------------|----------------------------------|
| <b>463</b> | <b>683</b>              | <b>706</b>              | <b>409</b> | <b>419</b> |                              | Catalog Number                   |
| 2,000      | –                       | –                       | –          | –          |                              | <b>Amphetamines</b>              |
| –          | 750                     | 1,250                   | 375        | 625        | 500                          | d-Amphetamine                    |
| –          | –                       | –                       | 375        | 625        | 500                          | d-Methamphetamine                |
|            |                         |                         |            |            |                              | MDMA                             |
| 1,000      | 150                     | 375                     | –          | –          |                              | <b>Barbiturates</b>              |
|            |                         |                         |            |            |                              | Secobarbital                     |
| –          | –                       | –                       | –          | –          |                              | <b>Benzodiazepines</b>           |
| 1,000      | –                       | –                       | –          | –          |                              | Lormetazepam                     |
| –          | 100                     | 300                     |            |            |                              | Nordiazepam                      |
|            |                         |                         |            |            |                              | Oxazepam                         |
| 125        | 37.5                    | 65                      | 37.5       | 65         | 50                           | <b>Cannabinoids</b>              |
|            |                         |                         |            |            |                              | 11-Nor-Δ-9-THC-9-COOH            |
| 500        | 225                     | 375                     | 113        | 185        | 150                          | <b>Cocaine</b>                   |
| 250        | 40                      | 70                      | –          | –          |                              | Benzoyllecgonine                 |
| 2.0        | 0.3                     | 0.8                     | –          | –          |                              | Ethanol (mg/dL)                  |
| 750        | 225                     | 375                     | –          | –          |                              | Lysergic Acid Diethylamide (LSD) |
| 750        | 225                     | 375                     | –          | –          |                              | Methadone                        |
|            |                         |                         |            |            |                              | Methaqualone                     |
| 750        | 225                     | 375                     | 1,500      | 2,500      | 2,000                        | <b>Opiates</b>                   |
| 200        | 19                      | 31                      | 19         | 31         | 25                           | Morphine (Free)                  |
| 750        | 225                     | 375                     | –          | –          |                              | Phencyclidine (PCP)              |
|            |                         |                         |            |            |                              | Propoxyphene                     |
| 2,000      | –                       | –                       | –          | –          |                              | <b>Tricyclic Antidepressants</b> |
|            |                         |                         |            |            |                              | Nortriptyline                    |

\*United States Substance Abuse and Mental Health Services Administration (formerly NIDA)  
Refer to the package insert for performance and stability claims.

# Liquichek Urine Toxicology Screen Controls



## Liquichek Urine Toxicology Control, Levels S1 & S2

A human urine control containing drugs of abuse at concentrations 20–25% above or below immunoassay screen cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
2,000 ng/mL  
Opiate cutoff



## Liquichek Urine Toxicology Control, Levels S1 Low Opiate & S2 Low Opiate

A human urine control containing drugs of abuse at concentrations 20–25% above or below immunoassay screen cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
300 ng/mL  
Opiate cutoff



## Liquichek Urine Toxicology Control, Level S3

A human urine control containing drugs of abuse at elevated levels for immunoassay screening procedures.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
300 ng/mL  
Opiate cutoff



## Liquichek Urine Toxicology Negative Control

Tested as negative for 32 of the most commonly abused drugs.

- 3 year shelf life at 2–8°C
- 60 day open-vial stability when stored refrigerated at 2–8°C
- Use with screening or confirmatory methods



## Liquichek Urine Toxicology Control, Levels S1S & S2S

A human urine control containing six drugs of abuse at concentrations 20–25% above or below SAMHSA's\* screen cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C
- Targets the 500 ng/mL Amphetamine and 150 ng/mL Cocaine cutoffs
- 2,000 ng/mL Opiate cutoff
- Includes MDMA

▶ For use with the 2,000 ng/mL Opiate, 500 ng/mL Amphetamine and 150 ng/mL Cocaine cutoffs.

# Additional Urine Toxicology Controls



## Liquichek Opiate Control

A human urine control containing eight drugs or drug metabolites in two levels, at concentrations above and below most assay cutoffs. This control has been tested by GC/MS, LC/MS/MS and other methods to confirm the recovery of drug quantities added, with representative values provided for screening and confirmatory methods.

- 3 year shelf-life at -20°C to -70°C
- 30 day open-vial stability at 2–8°C

### Analytes

6-Monoacetylmorphine  
(6-MAM; Heroin Metabolite)  
Buprenorphine

EDDP  
Fentanyl  
Meperidine\*\*

Methadone  
Oxycodone  
Tramadol

\*United States Substance Abuse and Mental Health Services Administration (formerly NIDA)

\*\*Not for diagnostic use in the United States.

Refer to the package insert for performance and stability claims.



## Liquichek Urine Toxicology Confirmatory Controls



### Liquichek Urine Toxicology Control, Level C1

A human urine control containing drugs of abuse at very low levels or at the Limits of Quantitation (LOQ) for confirmatory test methods.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
300 ng/mL  
Opiate cutoff



### Liquichek Urine Toxicology Control, Levels C2 Low Opiate & C3 Low Opiate

A human urine control containing drugs of abuse at concentrations 20–25% above or below confirmatory cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ For use with  
300 ng/mL  
Opiate cutoff



### Liquichek Urine Toxicology Control, Levels C2 & C3

A human urine control containing drugs of abuse at concentrations 20–25% above or below confirmatory cutoff levels.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C



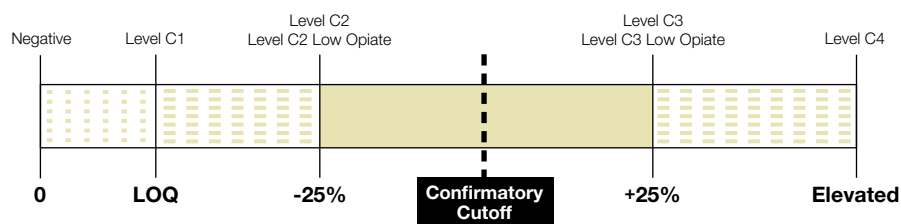
### Liquichek Urine Toxicology Control, Level C4

A human urine control containing drugs of abuse at elevated levels for monitoring confirmatory testing methods.

- 3 year shelf life at 2–8°C
- 30 day open-vial stability at 2–8°C

▶ Levels C2, C3 and C4 are designed for use with 2,000 ng/mL Opiate, 250 ng/mL Amphetamine and 100 ng/mL Benzoyllecgonine cutoff concentrations.

# Which Confirmatory Controls are Right for You?



## Liquichek Urine Toxicology Confirmatory Control Gravimetric Reference Values

| Analytes (ng/mL)                              | Level C1   | Level C2<br>Low Opiate | Level C3<br>Low Opiate | Level C2   | Level C3   | Level C4   |
|-----------------------------------------------|------------|------------------------|------------------------|------------|------------|------------|
| <b>Catalog Number</b>                         | <b>441</b> | <b>468</b>             | <b>469</b>             | <b>442</b> | <b>443</b> | <b>444</b> |
| <b>Amphetamines</b>                           |            |                        |                        |            |            |            |
| d-Amphetamine                                 | 100        | 375                    | 625                    | 188        | 312        | 750        |
| d-Methamphetamine                             | 100        | 375                    | 625                    | 188        | 312        | 750        |
| 3, 4-Methylenedioxymethamphetamine (MDMA)     | 100        | –                      | –                      | 188        | 312        | 750        |
| 3, 4-Methylenedioxyamphetamine (MDA)          | 100        | –                      | –                      | 188        | 312        | 750        |
| 3, 4-Methylenedioxy-N-ethylamphetamine (MDEA) | 100        | –                      | –                      | 188        | 312        | 750        |
| <b>Barbiturates</b>                           |            |                        |                        |            |            |            |
| Amobarbital                                   | 80         | 150                    | 250                    | 150        | 250        | 1,200      |
| Butalbital                                    | 80         | 150                    | 250                    | 150        | 250        | 1,200      |
| Pentobarbital                                 | 80         | 150                    | 250                    | 150        | 250        | 1,200      |
| Phenobarbital                                 | 80         | 150                    | 250                    | 150        | 250        | 1,200      |
| Secobarbital                                  | 80         | 150                    | 250                    | 150        | 250        | 1,200      |
| <b>Benzodiazepines</b>                        |            |                        |                        |            |            |            |
| Alpha-Hydroxyalprazolam                       | 80         | 225                    | 375                    | 225        | 375        | 1,000      |
| Nordiazepam                                   | 80         | 225                    | 375                    | 225        | 375        | 1,000      |
| Oxazepam*                                     | +          | +                      | +                      | +          | +          | +          |
| <b>Cannabinoids</b>                           |            |                        |                        |            |            |            |
| 11-Nor- $\Delta$ -9-THC-9-COOH                | 6          | 11.5                   | 18.5                   | 11.5       | 18.5       | 150        |
| <b>Cocaine</b>                                |            |                        |                        |            |            |            |
| Benzoylcegonine                               | 40         | 115                    | 185                    | 75         | 125        | 300        |
| Ethanol (mg/dL)                               | 20         | 40                     | 70                     | 40         | 70         | 250        |
| Lysergic Acid Diethylamide (LSD)              | 0.2        | 0.35                   | 0.65                   | 0.35       | 0.65       | 4.0        |
| <b>Methadone</b>                              |            |                        |                        |            |            |            |
| Methadone                                     | 120        | 225                    | 375                    | 225        | 375        | 1,000      |
| EDDP (Methadone Metabolite)*                  | +          | +                      | +                      | +          | +          | +          |
| Methaqualone                                  | 120        | 225                    | 375                    | 225        | 375        | 1,000      |
| <b>Opiates</b>                                |            |                        |                        |            |            |            |
| 6-Monoacetylmorphine*                         | +          | +                      | +                      | +          | +          | +          |
| Codeine                                       | 120        | 225                    | 375                    | 1,500      | 2,500      | 4,000      |
| Morphine-3- $\beta$ -d-Glucuronide            | 120        | 225                    | 375                    | 1,500      | 2,500      | 4,000      |
| Phencyclidine (PCP)                           | 10         | 19                     | 31                     | 19         | 31         | 100        |
| <b>Propoxyphenes</b>                          |            |                        |                        |            |            |            |
| Norpropoxyphene                               | 120        | 225                    | 375                    | 225        | 375        | 1,000      |
| Propoxyphene                                  | 120        | 225                    | 375                    | 225        | 375        | 1,000      |

\*Analyte present, no claims are made regarding performance or stability.  
Refer to the package insert for performance and stability claims.

## Ordering Information

| Cat #                                      | Product Name                   | Quantity   |
|--------------------------------------------|--------------------------------|------------|
| <b>Liquichek Urine Toxicology Controls</b> |                                |            |
| <b>Screen Controls</b>                     |                                |            |
| 461                                        | Level S1 . . . . .             | 10 x 10 mL |
| 462                                        | Level S2 . . . . .             | 10 x 10 mL |
| 466                                        | Level S1 Low Opiate . . . . .  | 10 x 10 mL |
| 467                                        | Level S2 Low Opiate . . . . .  | 10 x 10 mL |
| 463                                        | Level S3 . . . . .             | 10 x 10 mL |
| 409                                        | Level S1S . . . . .            | 10 x 10 mL |
| 419                                        | Level S2S . . . . .            | 10 x 10 mL |
| 423                                        | Level S1E Low Opiate . . . . . | 10 x 10 mL |
| 424                                        | Level S2E Low Opiate . . . . . | 10 x 10 mL |
| 438                                        | Level S1E . . . . .            | 10 x 10 mL |
| 439                                        | Level S2E . . . . .            | 10 x 10 mL |
| 683                                        | Level S1O Low Opiate . . . . . | 10 x 10 mL |
| 706                                        | Level S2O Low Opiate . . . . . | 10 x 10 mL |

### Confirmatory Controls

|     |                               |            |
|-----|-------------------------------|------------|
| 441 | Level C1 . . . . .            | 10 x 20 mL |
| 442 | Level C2 . . . . .            | 10 x 20 mL |
| 443 | Level C3 . . . . .            | 10 x 20 mL |
| 444 | Level C4 . . . . .            | 10 x 20 mL |
| 468 | Level C2 Low Opiate . . . . . | 10 x 20 mL |
| 469 | Level C3 Low Opiate . . . . . | 10 x 20 mL |

### Negative Control for Screening or Confirmatory Testing

|     |                           |            |
|-----|---------------------------|------------|
| 460 | Negative Control. . . . . | 10 x 20 mL |
|-----|---------------------------|------------|

Minipaks also available

| Cat #                                            | Product Name                              | Quantity |
|--------------------------------------------------|-------------------------------------------|----------|
| <b>Other Urine Toxicology Controls Available</b> |                                           |          |
| <b>Liquichek Qualitative Urine Control</b>       |                                           |          |
| 454                                              | Negative Level . . . . .                  | 6 x 3 mL |
| 455                                              | Positive Level . . . . .                  | 6 x 3 mL |
| 454X                                             | Bilevel MiniPak (1 of each level) . . . . | 2 x 3 mL |

### Liquichek Opiate Control

|          |                           |          |
|----------|---------------------------|----------|
| 12005786 | Level 1 . . . . .         | 6 x 5 mL |
| 12005787 | Level 2 . . . . .         | 6 x 5 mL |
| 12002594 | Level 1 MiniPak . . . . . | 1 x 5 mL |
| 12002595 | Level 2 MiniPak . . . . . | 1 x 5 mL |

To determine which controls best suit your assay requirements, please contact your local Bio-Rad office.



These controls are featured in the Unity Interlaboratory Program. Participation in the Unity Interlaboratory Program can help improve the quality of your laboratory test results and help you meet regulatory and accreditation requirements worldwide. You may also subscribe to one of our Unity QC Data Management Solutions designed to improve the effectiveness of your statistical process control. Ask your Bio-Rad Account Representative for more information.

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